Drift to Default: Long Autoregressive Video Rollouts Relax to a Prompt-Set State

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Abstract

Autoregressive video diffusion models generate long videos chunk by chunk, conditioned on their own output, and drift over tens of seconds. We show that in models that read their past through attention, this drift is *relaxation*: the colour and texture statistics of a rollout move from those of its opening toward a default, and the long-run state is an equilibrium,

$$x_\infty \approx \alpha\, x_{\text{open}} + (1-\alpha)\,\mu.$$

- **Generality.** We shift the brightness and colour of the same opening and measure how much survives. In 13 configurations of five open models (Self Forcing with DMD, SiD and GAN objectives, CausVid, LongLive, Rolling Forcing), the share α is between 0.02 and 0.67 at 31 s. The share depends strongly on the training objective even when the architecture is identical.
- **The default.** The default μ is where a fresh rollout of the same prompt ends ($r = 0.93$ across configurations). It moves with the lighting described in the prompt (16–45 L^*) and differs between models for the same words.
- **An exception.** A diffusion-forcing model that continues by re-denoising overlapping frames (SkyReels-V2) does not relax ($\alpha \approx 1$), whatever its guidance, history noise or number of steps.
- **Mechanism.** An attention sink keeps a fixed share of the opening for at least 120 s; without it the opening is forgotten. A linear chunk-recursion model fitted to these curves recovers the sink's per-chunk weight that we measure directly by bit-exact KV-cache replay (≈ 0.05). A tiny weight still sets the equilibrium because the prompt's per-chunk pull is equally small.
- **Two levers.**
  - *Raise α:* a training-free, content-aware correction increases α (+0.12), reduces drift, and keeps 83–93% of legitimate brightness changes; post-hoc colour matching keeps 7–12%.
  - *Move μ:* a lighting description calibrated to the opening's measured brightness reduces drift by 26% on Self Forcing; a free-form VLM caption does not.

Every comparison is paired by seed on one GPU architecture. Seven analyses were preregistered with hashed code, and we report all their outcomes, including the three primary hypotheses or studies that failed.

1 Introduction

Causal (autoregressive) video diffusion models [1, 2] make streaming, open-ended generation possible and underlie interactive world models. Their known weakness is drift: over tens of seconds the picture darkens, saturates, loses texture or collapses. Many recent methods target this failure, through rolling windows, attention sinks, re-aligned positions, longer self-rollouts and test-time correction [7, 8, 9, 12, 13]. They are usually evaluated with one sample per prompt, an aggregate quality score and no uncertainty. What drift actually *is* has not been measured.

We treat drift as a measurement problem and borrow the design tools of clinical research: paired seeds, controlled interventions, held-out prompts, cluster-bootstrap intervals and preregistered analyses. Our central finding is simple. A long rollout *relaxes*: every low-level statistic moves from its opening value toward a default set by the prompt, and a sink holds back a fixed share of the opening. This one picture explains several observations that otherwise look unrelated:

- why sinks reduce drift;
- why real camera frames "drift more" than generated ones;
- why a test-time correction that copies the anchor's tone works but over-corrects at early denoising steps;
- why describing the opening in the prompt fails while calibrating the lighting words succeeds.

Contributions.

1. A measurement protocol for long-video drift: self-referenced metrics, held and *perturbed* openings, paired seeds pinned to one GPU architecture, and preregistration. It also exposes three evaluation pitfalls:
   - the GPU architecture changes bf16 outputs;
   - resumed jobs silently break pairing;
   - a standard image-quality metric is blind to drift.
2. The relaxation law, $x_\infty \approx \alpha x_{\text{open}} + (1-\alpha)\mu_{\text{prompt}}$. We establish it interventionally, by shifting the brightness and colour of the same opening, across 13 configurations of five models, with an exception for diffusion forcing that points to the conditioning mechanism.
3. A quantitative mechanism. Long-horizon persistence curves and direct KV-cache interventions give consistent per-chunk weights, which explains how a sink with a tiny weight sets the equilibrium.
4. Two practical levers matching the two terms of the law, both evaluated in preregistered tests against baselines: a content-aware correction raises α, and calibrated prompts move μ.

No model is trained. Everything is inference on public weights, so the protocol applies to any open autoregressive video model.

2 Related work

Autoregressive video diffusion.

- Diffusion Forcing [3] trains with independent per-frame noise.
- CausVid [1] distils a bidirectional teacher [4] into a causal few-step student with DMD [5].
- Self Forcing [2] trains on the model's own rollouts.
- Long-video successors add rolling KV windows, attention sinks (as in streaming language models [11]), RoPE re-alignment, LoRA-based extension, and joint rolling denoising [7, 8, 9].
- SkyReels-V2 [10] extends videos by diffusion forcing over overlapping segments.

Error accumulation and correction. Meta-ARVDM [14] analyses error accumulation theoretically. Pathwise test-time correction [12] and FreqForcing [13] correct drift at inference; our correction is a deliberately minimal member of this family. It transfers only the DC tone at one early step, which lets us study the step as a knob.

Evaluation. VBench [15] scores imaging quality with MUSIQ [16]. We show that MUSIQ is nearly blind to drift, and we measure drift relative to each prompt's own fresh-start distribution, in colour statistics and in DINOv2 [17] features.

Table 1: Model configurations. All except SkyReels share the Wan2.1-T2V-1.3B architecture and are run in our harness (`hold_gen.py`, bit-identical to the official Self Forcing loop when no perturbation is applied); window and sink sizes are in latent frames (3 per chunk of 0.75 s).

Configuration	Training / generation	Steps	Context
Self Forcing DMD	self-rollout, DMD [5]	4	window 21; sink 0 / 1 / 3
Self Forcing SiD, SiD v2	self-rollout, SiD [6]	4	window 21, sink 3
Self Forcing GAN	self-rollout, GAN	4	window 21, sink 3
Self Forcing 10 s	self-rollout on 10-s clips	4	window 21, sink 3
CausVid [1]	teacher-forced DMD	3	window 21, sink 3
LongLive [7]	long-rollout LoRA	4	window 12; sink 0 / 3
Rolling Forcing [8]	rolling joint denoising	5	anchor block + 21
SkyReels-V2-DF [10]	diffusion forcing, 544p/24 fps	30 + CFG	overlap 17 frames

3 Setup

Models. All configurations generate 832×480 video at 16 fps in chunks of 3 latent frames (0.75 s), except SkyReels (960×544, 24 fps); details are in Table 1.

- **Self Forcing [2] checkpoints:** DMD (main), SiD, SiD-v2, GAN, and the 10-second variant. All share one architecture (Wan2.1-1.3B [4]) and differ only in the distillation objective.
- **CausVid [1]:** autoregressive DMD checkpoint.
- **LongLive [7]:** LoRA, official window 12 + sink 3.
- **Rolling Forcing [8]:** rolling joint denoising with a fixed anchor block.
- **SkyReels-V2-DF-1.3B [10]:** diffusion forcing over overlapping segments.

Held and perturbed openings. To separate drift from what the prompt asks for, an opening image is held for 0.56 s, encoded as the first chunk (or as the prefix video for SkyReels), and continued for 31 s with a generic driving prompt. Openings are NAVSIM [19] front-camera frames or first frames generated by the model. In the interventional study, the same opening is shifted in CIELAB before encoding: L^* by ± 8 and ± 16, and b^* by ± 12.

Metrics. For each video we record per-frame Lab means, saturation, L^* contrast and Laplacian sharpness. The *persistence* α is the within-scene slope of the end value (last 2 s) on the opening value. For prompt-only rollouts, drift is measured against 10 fresh 5-second generations of the same prompt: D is the Mahalanobis distance of colour statistics and K the DINOv2 nearest-neighbour distance.

Statistics. Seeds are paired across arms, and every arm of a scene runs in one job on one GPU architecture. We use mixed models or seed-averaged paired differences, with scene-cluster bootstrap intervals and Holm correction within each preregistered family. Each preregistration froze its code by SHA-256 hash before generation, and the last four were committed publicly first.

4 Drift is relaxation toward a prompt-set default

Observational evidence. Across 40 held openings with a generic prompt, the change of every statistic over 31 s is strongly anti-correlated with its opening value (Table 2): r between -0.79 and -0.95 for L^*, a^*, b^*, saturation, sharpness and contrast, on both Self Forcing and LongLive. For brightness:

$$\Delta L = 22.8 - 0.69\, L_{\text{open}}, \quad R^2 = 0.82.$$

- Only 22–40% of an opening's deviation survives.

Table 2: Every low-level statistic relaxes toward a default (40 held openings, generic prompt). Persistence = 1 + slope of the 31-s change on the opening value; r = correlation of change with opening value.

Statistic	Self Forcing persistence	r	LongLive persistence	r
L*	0.31	-0.91	0.26	-0.81
a^*	0.38	-0.92	0.39	-0.90
b^*	0.22	-0.88	0.42	-0.80
saturation	0.54	-0.92	0.23	-0.95
log sharpness	0.37	-0.91	0.23	-0.94
L* contrast	0.34	-0.82	0.37	-0.79

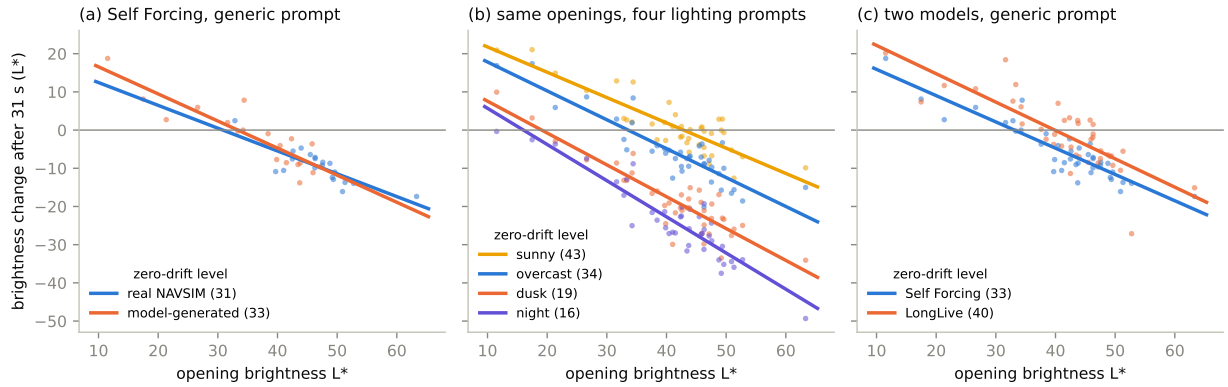


Figure 1: Brightness change after 31 s against the opening's brightness (one dot per opening). (a) Real and generated openings follow one line. (b) Four lighting prompts set four defaults. (c) Two models: same slope, different default.

- Real and generated openings fall on the same line ($p = 0.80$), so real frames "drift more" only because they are brighter.

The default is set by the prompt. Continuing the same openings under four lighting prompts gives four levels (Fig. 1b). On Self Forcing these are sunny 43, overcast 34, dusk 19 and night 16 L*; the lighting term has $p \approx 5 \times 10^{-78}$. On LongLive the same words give 45, 40, 32 and 21. The prompt sets the target, and the mapping from words to brightness is model-specific.

Interventional evidence across models (preregistered). Observational regressions compare different scenes, whose content differs along with their brightness. We therefore shifted the opening *within* each of 24 scenes (12 NAVSIM, 12 generated): L* by $-16, -8, 0, +8, +16$ and b^* by ± 12. Each was continued for 31 s by 13 configurations (Table 1); every configuration of a scene ran in one job on one GPU architecture. The persistence α is the within-scene slope of the end value on the opening value. Fig. 2a shows the results.

- **Every attention-based model relaxes.** α_L ranges from 0.02 (LongLive without a sink) to 0.67 (Self Forcing GAN); colour temperature behaves alike.
- **The training objective matters even with everything else fixed.** Five Self Forcing checkpoints share architecture, context and sampler, yet α_L ranges from 0.12 (SiD) to 0.67 (GAN) and the default from 17 to 47 L*.
- **The default is the end of a fresh rollout.** Across 10 configurations, the zero-drift level estimated from the shifted openings correlates $r = 0.93$ with where prompt-only fresh starts end (Fig. 3). A model's default can therefore be measured from a few cheap fresh rollouts.
- **Diffusion forcing is the exception.** SkyReels-V2-DF keeps the opening: $\alpha_L = 1.09$ $[1.02, 1.16]$ at 10 s and 0.96 $[0.86, 1.07]$ at 20 s (12 scenes). In an exploratory check, removing classifier-free guidance,

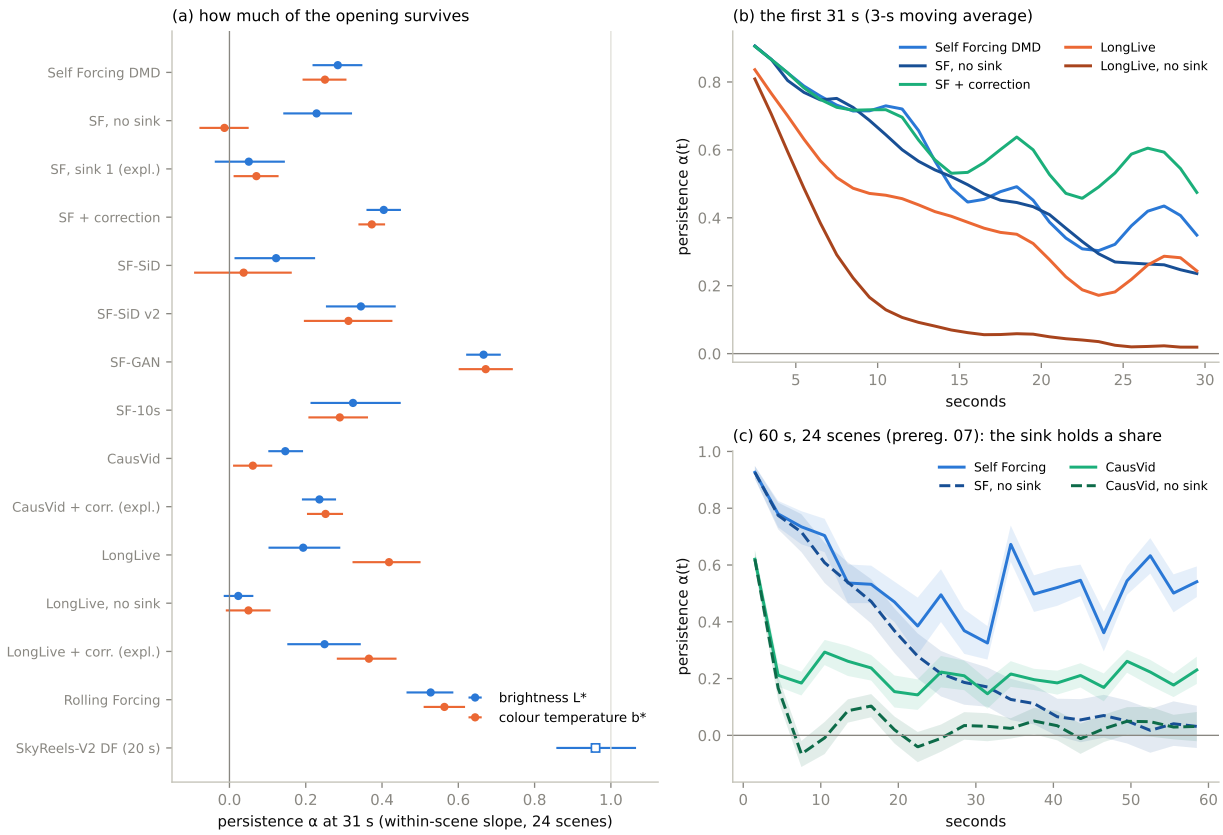


Figure 2: Relaxation study (preregistration 06). (a) Persistence of a within-scene shift of the opening at 31 s (24 scenes; SkyReels at 20 s, 12 scenes, hollow marker). (b) Persistence over time for the sink and correction contrasts. (c) Preregistration 07 (24 scenes, 60 s; bands are scene-bootstrap 95% CIs): with a sink a fixed share of the opening is held; without one it is forgotten.

tripling the history noise, or cutting to 10 steps all left $\alpha \approx 1$. Its continuation re-denoises 17 overlapping frames, a pixel-level constraint, whereas the KV-cache models only attend to their past. We conjecture that the conditioning mechanism, not autoregression itself, decides whether a model relaxes.

5 Mechanism: an equilibrium between the anchor and the prompt

Model a chunk's mean statistic as a linear recursion:

$$x_{k+1} - \mu = w_w\big(\bar{x}_{\text{window}} - \mu\big) + w_s\big(x_{\text{sink}} - \mu\big).$$

The opening counts as a window chunk until it rolls out; after that, with a sink, it keeps a direct pull w_s. The prompt's pull is $w_p = 1 - w_w - w_s$. Without a sink the opening is eventually forgotten ($\alpha \to 0$). With a sink the equilibrium share is

$$\alpha_\infty = \frac{w_s}{w_s + w_p}.$$

The share is large whenever the prompt's pull is as weak as the sink's.

Two independent measurements agree. Bit-exact KV-cache replay lets us shift only the sink or only the window and regenerate the next chunk from identical noise. It measures $w_s = 0.053$ $[0.040, 0.066]$ for synthetic tone shifts, 0.036 in a dose-response design, and $w_w \approx 0.76$ (synthetic) to 0.90–0.99 (natural differences). Fitting the recursion to the long-horizon persistence curve alone gives:

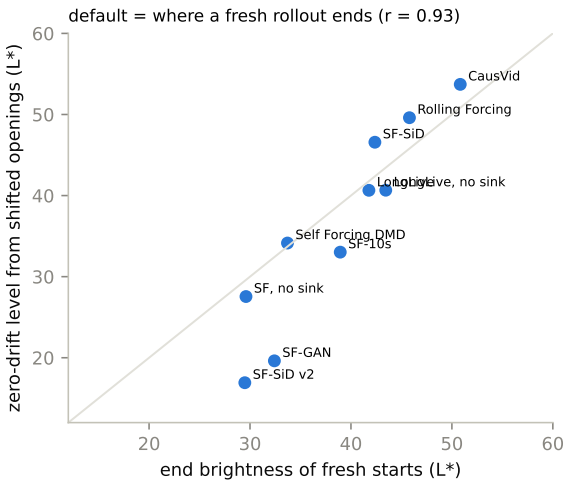


Figure 3: The zero-drift level from shifted openings equals the end brightness of fresh rollouts of the same prompt.

- Self Forcing: $w_s = 0.049$ $[0.005, 0.069]$, $w_w = 0.85$, $\alpha_\infty = 0.32$;
- LongLive: $w_s = 0.054$, $w_w = 0.81$.

This resolves an earlier puzzle: the sink's direct effect on the next chunk is tiny, yet it roughly halves drift.

Interventional tests (preregistered, Holm over four).

- **LongLive.** Removing the sink lowers α_L from 0.19 to 0.02 (-0.17 [-0.28, -0.06], $p_{\text{Holm}} = 0.013$).
- **Self Forcing, colour.** Removing the sink lowers α_b from 0.25 to -0.01 (-0.26 [-0.34, -0.18], $p < 0.001$).
- **Self Forcing, brightness: not supported at 31 s** (-0.06 [-0.17, 0.05]). Its window forgets slowly ($w_w \approx 0.9$), so at 31 s the sink-less run had not yet forgotten. In the preregistered long-horizon subset (6 scenes, 120 s), the sink holds 0.44–0.54 from 60 s on, while without it persistence is -0.10 to -0.07. The difference is $+0.56$ $[0.34, 0.75]$ at 118 s.

Confirmation at 60 s (preregistration 07). We then tested the sink at 60 s on all 24 scenes, for Self Forcing and for CausVid (288 videos, Fig. 2c).

- **Self Forcing:** with a sink, 0.58 of the opening remains; without one, 0.03. Difference $+0.55$ $[0.45, 0.65]$, $p_{\text{Holm}} < 10^{-9}$.
- **CausVid:** 0.23 vs 0.04, difference $+0.19$ $[0.14, 0.24]$, $p_{\text{Holm}} < 10^{-6}$.
- **Quantitative predictions.** Without a sink, Self Forcing's persistence fell from 0.24 at 30 s to 0.03 at 60 s, as predicted. With a sink, the plateau of 0.58 lies just above the band the recursion predicted, $[0.25, 0.54]$: the model gets the shape right and the level slightly low.

Fitting the recursion. Fitted to the 120-s curve, the model gives:

- Self Forcing: $w_s = 0.060$ $[0.021, 0.097]$, $w_w = 0.86$, $\alpha_\infty = 0.42$ $[0.25, 0.54]$. KV-cache replay measured 0.053 $[0.040, 0.066]$.
- LongLive: $w_s = 0.034$ $[0.027, 0.045]$.
- Ordering on the 31-s curves: the fitted sink weights follow the order of α (GAN 0.11, SiD and Rolling Forcing 0.08, CausVid 0.05, LongLive 0.02).

Table 3: Correction versus post-hoc colour matching on the preregistered confirmation set (40 held-out prompts × 4 seeds; transitions: 8 held-out prompts with legitimate brightness changes).

Method	ΔD [95% CI]	Δz_L	transitions kept	jitter
Content-aware correction, $\beta = 1$	-1.48 [-1.76, -1.19]	+1.08	0.83	×1.13
Content-aware correction, $\beta = 1.5$	-1.78 [-2.11, -1.45]	+1.32	0.93	×1.67
Post-hoc mean matching	-0.89 [-1.29, -0.50]	+1.55	0.07	×0.15
Post-hoc Reinhard [20]	-1.58 [-1.96, -1.19]	+1.53	0.12	×0.13

Two caveats:

- From a 31-s curve alone, Self Forcing's sink weight is not identifiable, because the plateau is not reached yet.
- Both long curves dip near 30 s and then recover, which the first-order model does not capture. With six scenes this may be noise.

6 Two levers

Raising α: a content-aware correction. At denoising step 1 of 4, the noisy chunk is run once more with a temporary KV cache that holds only the anchor. The difference between the per-channel spatial means (the DC tone) of that prediction and of the real one is added back, scaled by β. Nothing else changes, including the main cache and the random stream.

- **Confirmation (preregistered, 40 held-out prompts × 4 seeds).** Colour drift D falls by 1.48 $[1.19, 1.76]$ at $\beta = 1$ and by 1.78 at $\beta = 1.5$. Structure drift K is non-inferior. On held-out prompts with legitimate brightness changes, 83% / 93% of the change is kept.
- **Baselines.** Post-hoc colour matching to the anchor (mean, or Reinhard mean and standard deviation) reaches similar D (-0.89, -1.58) but keeps only 7% / 12% of legitimate transitions (Table 3).
- **The step is a knob.** Correcting at step 0 removes more drift but erases tunnels (56% kept); steps 2–3 do nothing.
- **Cost and transfer.** The correction costs one forward pass per chunk and some flicker. On LongLive it reduces D by half as much.
- **It works by raising α.** In the relaxation study the correction raises Self Forcing's persistence from 0.28 to 0.41 (+0.12 $[0.05, 0.20]$, preregistered). In exploratory runs it does the same on CausVid (+0.09 $[0.03, 0.14]$), and on LongLive the estimate is +0.06 [-0.06, 0.18]. It is therefore an artificial strengthening of the sink's pull, restricted to the DC tone.

Moving μ: calibrated prompts.

- **Free-form caption (preregistered).** Describing the opening with a Qwen2.5-VL [18] caption does not reduce brightness drift on either model (Self Forcing -1.21 [-3.09, 0.36]). The caption describes half of the normally lit frames as "dim" and pulls them darker. It does improve content retention (+0.10 / +0.15).
- **Calibrated lighting (preregistered).** Choosing the lighting sentence whose measured default is nearest the opening's brightness reduces $|\Delta L|$ by 26% on Self Forcing (Holm $p = 0.013$); with the correction, by 44%.
- **Calibration is per model.** On LongLive, the preregistered test was not significant even with LongLive's own calibration; the lighting-only prompt reduces drift by 34% (secondary).

(Fig. 5).

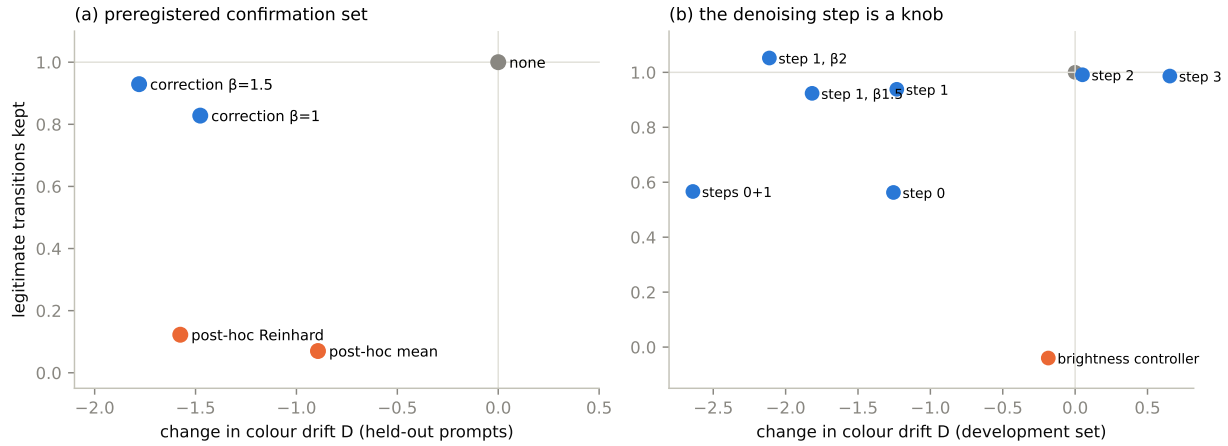


Figure 4: Drift reduction versus legitimate brightness transitions kept. (a) Preregistered confirmation set: only the content-aware correction is in the upper left. (b) Development set: the denoising step at which the anchor's tone is transferred trades drift removal against content fidelity.

The law predicts these outcomes out of sample. Take α and the generic default from the relaxation study (different scenes) and the lighting defaults from the lighting experiment (different openings). Then $\Delta L \approx (1-\alpha)(\mu - L_{\text{open}})$ predicts the 31-s brightness change of every opening under the generic and the calibrated prompt in the calibrated-lighting experiment:

- correlation 0.82 on Self Forcing and 0.77 on LongLive, mean absolute error 3.8 L*;
- the mean benefit of calibration is overpredicted (-4.5 predicted vs -3.2 and -2.8 observed).

The law is a usable forecast of how a prompt choice will change long-horizon brightness.

7 Context strategies

A preregistered factorial study (10 strategies, 40 prompts $\times$ 4 seeds) found:

- a 3-frame sink cuts colour drift by 7.56 $[6.81, 8.32]$;
- window size matters only without a sink;
- RoPE re-alignment, which Self Forcing was not trained for, does not help;
- MUSIQ varies only between 66 and 69 across strategies whose drift differs threefold;
- the GPU architecture changes outputs, so 70% of nominally paired comparisons were not paired.

8 Discussion: what this means for long-horizon world models

- **Drift is a forgetting-and-reverting process, not accumulating noise.** In every attention-based model we tested, a rollout forgets its opening at a model-specific rate and reverts to the default its prompt sets. The default is cheap to measure from a few fresh rollouts.
- **Two measurable numbers describe a model's long-horizon fidelity:** the share of the opening it keeps (α) and its default (μ). Both change with the training objective even when architecture and inference are fixed. Reporting them would make long-video models comparable in a way that a single quality score is not.
- **The levers follow from the law.**

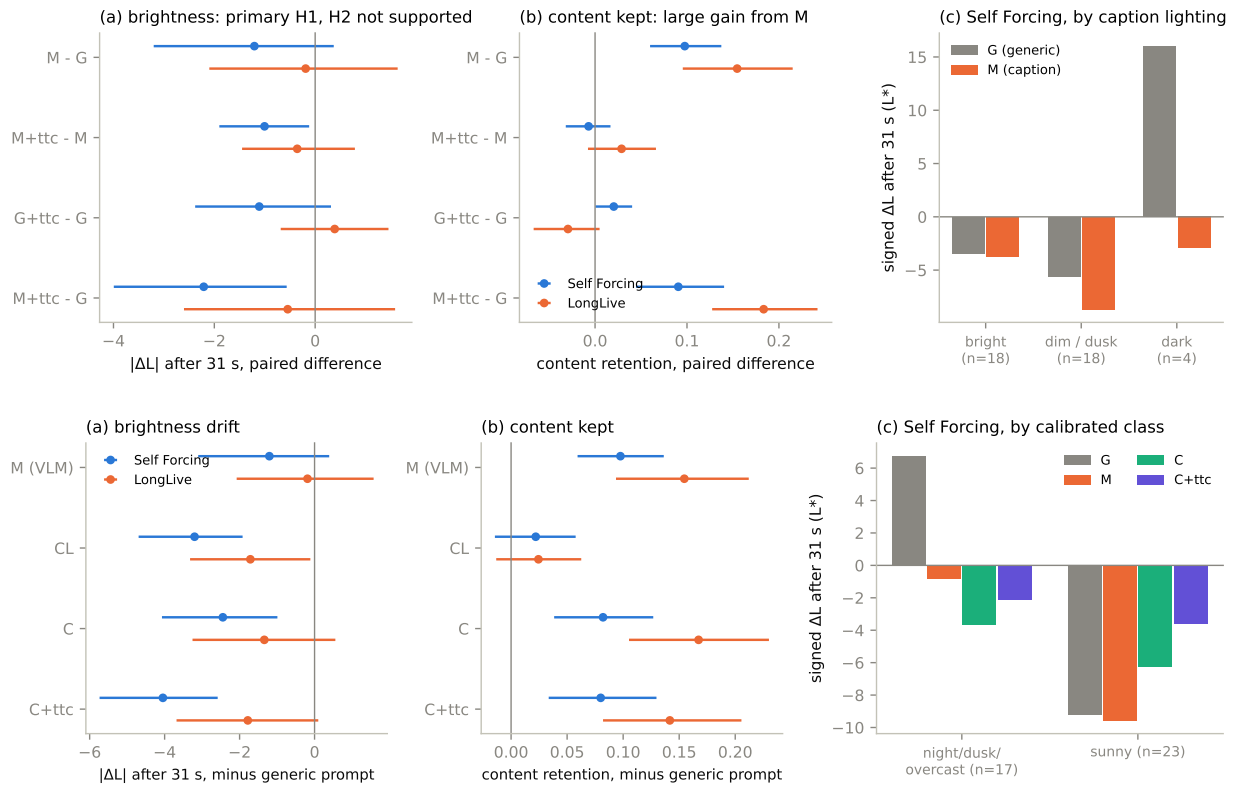


Figure 5: Moving μ. Top: free-form VLM captions (preregistered; not supported for brightness, large content gain). Bottom: brightness-calibrated lighting descriptions (preregistered; supported against the generic prompt on Self Forcing). Paired differences with opening-cluster bootstrap 95% CIs.

  - Raise α: keep the opening as a sink, or transfer the anchor's tone at an early denoising step, which keeps legitimate changes because it acts only on the DC component.
  - Move μ: describe the lighting with words whose measured default matches the opening.

  A world model conditioned on a real camera frame and a caption will drift toward the caption unless the caption's default matches the frame.

- **Conditioning by overlap is different.** The diffusion-forcing model kept its opening. Whether this comes from re-denoising overlapping frames or from other differences between SkyReels and the KV-cache models is the most important open question here; we report it as an exception, not a mechanism.

9 What did not hold

We report claims that were revised or withdrawn after checks, and the preregistered predictions that failed.

- **Withdrawn claims:**
  - that MUSIQ rewards collapsed videos (it is merely insensitive);
  - that the sink stabilises the default (trend difference +0.5 [-2.4, 3.5]);
  - that the model follows its context by 75% (only for artificial shifts);
  - that early dynamics forecast the final brightness.
- **Failed preregistered tests:**

  - describing the opening with a free-form caption (study 3);
  - calibrated lighting on LongLive (studies 4 and 5);
  - the sink effect on Self Forcing brightness at 31 s (study 6). It is supported at 60 s in study 7, but that endpoint was chosen after seeing study 6.

- **A failed quantitative prediction:** the recursion predicted a Self Forcing plateau of 0.25–0.54, and the observed plateau was 0.58 (study 7).

- **Unexplained:** a sink of one latent frame retained less of the opening than no sink (exploratory).

10 Limitations

- All models but SkyReels share the Wan2.1-1.3B base, and only one diffusion-forcing model was tested.
- Horizons are 31–120 s; openings are held images, not real video, and are continued with one generic prompt in the interventional studies.
- The statistics are global colour and texture summaries. Structural failures (objects melting, geometry drift) are not captured by α.
- The correction acts on colour only. Lighting calibration uses four discrete levels and cannot match openings brighter than the sunny default.
- One seed per opening version in the interventional studies; precision comes from scenes and versions instead.

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A Preregistrations, deviations and pitfalls

Table 4: The seven preregistered studies. Code was frozen by SHA-256 hash before generation; studies 3–7 were also committed to the public repository before generation.

#	Study	Primary hypotheses	Outcome
1	Context strategies (2,000 videos)	sink lowers drift; window matters only without sink	supported
2	Content-aware correction (480 + 96 videos)	lower colour drift; structure non-inferior; transitions kept	supported
3	Caption prompts (800 videos)	caption < generic; +correction < caption	not supported
4	Calibrated lighting (560 videos)	calibrated < caption; calibrated < generic	supported vs generic (SF)
5	Calibrated lighting, LongLive's own levels	same, LongLive	not supported
6	Relaxation (2,184 + 36 videos)	sink raises α (two models); correction raises α; sink raises α_b	3 of 4 supported (SF brightness not at 31 s)
7	Sink retention at 60 s (288 videos)	sink raises α at 60 s (SF, CausVid)	supported (both)

Deviations.

- In study 3, a preempted job resumed on another GPU type and split four LongLive openings across architectures. We regenerated the affected arms on the original architecture; the conclusions are unchanged (sensitivity analysis in the repository).
- In study 6, the ODE-initialised student was replaced by SiD v2 before generation, because it collapses to blur after the opening.
- Exploratory configurations (sink 1; correction on LongLive and CausVid) were added after freezing and before any data, and are reported as exploratory.

Pitfalls for evaluating video models.

1. **The GPU architecture changes the video.** With bf16, the same seed and code give bit-identical videos within an architecture but different videos across A40, A100/A800 and L40S/RTX 6000 Ada. Paired designs must pin the architecture.
2. **Resumed jobs can silently break pairing.** Generators that skip existing outputs resume a preempted job wherever it lands.
3. **Popular quality metrics miss drift.** MUSIQ, used by VBench for imaging quality, barely moves across strategies that differ threefold in colour drift.

4. **Natural variation is too small to estimate persistence.** Within a prompt, seeds start only about 2 L^* apart, so within-prompt slopes of end on start brightness have confidence intervals of ± 0.4. Controlled shifts of ± 16 L^* are needed.

5. **Change-on-baseline regressions can mislead.** Regressing change on an opening value measured with error produces spurious regression to the mean (Oldham's problem). Held openings are measured almost without error, and the interventional design removes the issue.